

Who Works With Whom? Gender, Nationality, and the Shape of Co-Publishing Networks

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1 Introduction

- Why do co-publishing networks matter?
 - Co-authorship structures shape productivity, visibility, and citation impact, and they evolve systematically over time (Abbasi et al., 2011; Ductor, 2015; Li et al., 2022; Moody, 2004; Newman, 2001; Wallace et al., 2012)
 - Network mechanisms (such as: homophily, triadic closure, preferential attachment, structural holes) organize collaboration opportunities and inequalities (Barabasi & Albert, 1999; Bojanowski & Czerniawska, 2020; Ebadi & Schiffauerova, 2015)
- Which structural inequalities are perpetuated by the structure of co-publishing networks
 - To what extent and why are there gender differences in co-publishing?
 - * Women are underrepresented in high-status outlets and network positions persists across field (Bendels et al., 2018; Bravo-Hermsdorff et al., 2019; Bunker Whittington et al., 2024; Kim et al., 2022; Whittington, 2018).
 - * Research indicates gender assortativity and role-segregation, and topic-based homophily (Gao et al., 2022; Kretschmer et al., 2015).
 - * Outside academia, evidence of gendered networking gaps underscores the mechanism plausibility (Contreras et al., 2025).
 - To what extent and why are there nationality-based differences in co-publishing?

- * scholars preferentially collaborate with same-nationality peers, reflecting shared language, cultural understandings, and institutional ties that lower collaboration costs (Fuligni et al., 2005; Heiberger & Riebling, 2019; Ossenblok et al., 2014; Pillai et al., 2019).
- To what extent do gender and nationality-based differences in co-publishing intersect?
 - * what is intersectionality?
 - definition of intersectionality (Crenshaw, 1989; King, 1988).
 - * additive accounts understate disadvantage that accrues at the intersection of gender and nationality (Kozlowski et al., 2022).
 - * network status processes (visibility, centrality, gatekeeping, sponsorship) may compound for women from underrepresented nationality groups; conversely, shared nationality may buffer gender penalties in some contexts (Bunker Whittington et al., 2024; Kim et al., 2022; Whittington, 2018).
 - * Intersectional compounding vs. buffering: gender penalties may intensify for international women, or shared nationality may mitigate gender gaps in forming ties. (Kwiek & Roszka, 2021; van der Wal et al., 2021)
- What are potential underlying mechanisms?
 - * **Homophily**: same-gender and same-nationality ties lower collaboration costs and increase trust—predicting within-group clustering.
 - * **Status-seeking/bridging**: scholars (especially early-career) may pursue higher-status or internationally central alters (Bojanowski & Czerniawska, 2020; Ebadi & Schiffauerova, 2015; Eberle et al., 2021; Li et al., 2022).
- What are the empirical gaps this paper addresses?
 - * Abundant work on gender or nationality separately; far less on their intersection.
 - * Limited evidence on whether intersectional gaps arise from initiating vs. receiving ties, and whether effects vary by partner centrality or team span (cf. Bunker Whittington et al., 2024; Eberle et al., 2021).
 - * Studying gender, nationality differences and their intersection controlled for endogenous network effects, doing so while using temporal network models (Abbasi et al., 2011; Ferligoj et al., 2015)
- What are the research questions:
 - **RQ1**: Do men and women, and native and international scholars have different co-publishing network? For example, do they differ with regard to the number collaborators they have, how tightly their circles cluster, and how central they are in their network?
 - * Measures: network size and coverage, density, degree distribution, average path length, global/local clustering, transitivity, and assortativity (by gender, nationality, and intersec-

tional blocks), brokerage/bridging (betweenness, constraint), team-size and team-span diversity.

- **RQ2:** Do men and women, and native and international scholars, differ in how likely they are to start collaborations, and do they tend to choose collaborators who share their gender or nationality?
 - * Design: estimate homophily and status asymmetries with longitudinal tie-formation models including structural effects (closure/activity/popularity), and covariate effects for gender and nationality: egoX, altX, sameX.
- **RQ3:** When we consider gender and nationality together, do collaboration gaps widen for international women, or can sharing a nationality with potential partners help narrow those gaps?
- **RQ4:** Across gender $\times$ nationality groups, do researchers differ in how often they collaborate with more senior and more central collaborators? And how do a scholar’s own seniority and centrality shape these opportunities: do they amplify or reduce any gaps between groups?

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